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1 May 1978
SUPERSEDING
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MILITARY SPECIFICATION

RACKS, DELIVERY AND STORAGE

FOOD CONTAINER

This specification is approved for use by the US Army Natick Research and Development Command (GL), Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plated steel, delivery and storage racks.

1.2 Classification. Delivery and storage racks covered by this specification shall be as follows (see 6.2):

Type I - Flat shelves

Size 2 - 6 shelves
Size 4 - 8 shelves

Type II - Special shelves

Size 5 - V-top with drop leaf

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7330

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SPECIFICATIONS

FEDERAL

QQ-W-461 - Wire, Steel, Carbon, (Round, Bare and Coated)

MILITARY

MIL-B-2316 - Bakery Equipment, Including Units, Assemblies, Repair Parts and Tools, Preparation for Delivery Of

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-13-210 - Racks, Delivery, and Storage

(Copies of specifications, standards, and drawing required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM) STANDARD

A 519 - Seamless Carbon and Alloy Steel Mechanical Tubing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL SANITATION FOUNDATION (NSF)

Criteria C-2 Special Equipment and Devices

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, MI 48105.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. Delivery and storage racks delivered under this specification shall be the manufacturer's commercially current standard product, except for any changes necessary to comply with specification requirements.

3.2 Sanitary standard. The delivery and storage racks covered by this specification shall conform to the applicable requirements of NSF Standard C-2.

* 3.2.1 NSF compliance. Prior to approval of first shipment, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the racks he proposes to furnish under this specification meet the applicable requirements of NSF Standard No. C-2. Satisfactory evidence of this compliance shall be any of the following:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished rack, or

(2) A certification for the rack issued by NSF under this special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the racks are being procured, indicating that the racks have been tested and conform to NSF No. C-2.

3.3 Materials and components. Materials and components specified shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial delivery and storage racks. Commercial tolerances on dimensions shall apply (see 6.3).

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3.3.1 Steel.

* 3.3.1.1 Tube steel. Tube steel shall be seamless, cold worked, 1020 steel, conforming to ASTM A 519.

3.3.1.2 Steel rod. Steel rod shall conform to any steel numbers, finish 1 temper annealed of QQ-W-461.

* 3.4 Design and construction. The delivery and storage racks shall be constructed as specified herein and as illustrated in Drawing 5-13-210. The frame shall be all welded construction of 1-inch (25.4 mm) round or square tube steel as specified in 3.3.1.1. Racks shall show no evidence of parts breakage, separation or looseness and the shelves shall show no deflection greater than 1/8 of an inch (3.2 mm) when loaded and rolled over a 1/4-inch (6.4 mm) high obstacle when tested as specified in 4.4.1.

3.4.1 Shelves.

* 3.4.1.1 Wire shelves. Racks for shelves shall be constructed of 1/2-inch (13 mm) round steel rod specified in 3.3.1.2 recessed and welded into the frame. Frames for the shelves shall be constructed on 1-inch (25.4 mm) round tube steel specified in 3.3.1.1.

* 3.4.1.2 Drop leaf. The drop leaf for the type II rack shall be 10 $\pm$ 1/2 inch wide by 30 $\pm$ 1/2 inch long (254 $\pm$ 13 by 762 $\pm$ 13 mm), 1 inch (25.4 mm) thick plywood or partical board with the top and edges covered with 20 gauge (.035 inch - 0.9 mm) thick, type 302 or 304 stainless steel sheet with a No. 3 finish. The sheet shall be cemented to the board, rounded or finished smooth at the edges and corners. The shelf shall be fastened to the frame by a folding bracket (see 6.4) capable of supporting uniform distributed load of 50 pounds (23 kg) without permanent deformation or parts breakage.

* 3.4.2 Casters. The delivery and storage racks shall be provided with four casters. Casters shall be 5-inch (127 mm), stem type, swivel casters with a hard rubber tread 1-inch (25.4 mm) minimum width. Swivels shall be ball-bearing type and wheel bearings shall be ball or roller type. Wheel bearings and swivel bearings shall be permanently lubricated or shall be accessible for lubrication.

* 3.4.3 Bumpers. The rack shall be provided with bumpers so that no component of the rack, except the bumper, touches a vertical wall upon contract. The bumper shall be firmly assembled in accordance with the manufacturer's commerical practice and be capable of being easily removed and replaced when necessary. The bumper shall be fabricated from rubber or rubber-like material with a Durometer hardness of 75 $\pm$ 15.

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3.5 Finish. All metal components of the delivery and storage racks shall be chrome plated or epoxy coating in accordance with the manufacturer's standard practice. The underside of the drop leaf shall be painted to match the frame.

3.6 Workmanship. All metal forming shall be done with dies or fixtures to insure uniformity of size and shape. Bolts, nuts, and screws shall be drawn tight and secured so that they shall not work loose. Welding shall be free from oxide, scale, or grease. Welds shall be continuous and free from cracks, undercuts, burn through or deformation of material. All flux deposits shall be removed. All rough and sharp edges, burrs and slivers shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 In-process inspection. Visual and dimensional examination shall be made to determine that metal forming is as specified in 3.6. When nonconformance is noted, correction shall be made.

4.2.2.1 Intermediate testing. One bumper from each lot shall be tested for Durometer hardness as required in 3.4.3. Any nonconformance shall constitute failure of this test and the lot rejected.

4.2.3 End item inspection. The lot shall consist of all racks of one size and type offered for inspection at one time. The sample unit shall be one completely assembled rack.

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4.2.3.1 Visual examination. Examination shall be made of the racks for defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of visual defects

Examine	Defect	Classification Major	Minor
Finish	Not chrome plated or epoxy coated in accordance with manufacturer's standard practice	X	
	Rough or sharp edges, burrs, or slivers	X	
Design, construction and workmanship (general)	Component missing	X	
	Component split, fractured or bent	X	
	Component loose		X
	Not design specified	X	
	Casters not as specified	X	
	Bumper missing, not removable	X	
	Folding bracket not as specified		X
	Bracket does not support shelf or stow in down position	X	
	Welding missing or cracked	X	
	Welding not free from rust, scale, flux deposits, burns or undercut		X
Marking for identification	Missing, incomplete, illegible		X

4.2.3.2 Dimensional examination. Examination shall be made of the completed racks to determine compliance with dimensions specified. Any dimension not within the specified tolerance shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.3.3 End item testing. One rack from the production lot shall be tested as specified in 4.4.1 and 4.4.2, and when applicable 4.4.3 Any nonconformance shall be cause for rejection of the lot.

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4.2.4 Examination of preparation for delivery. Examination shall be made for preservation, packaging, packing, and marking in accordance with MIL-B-2316.

* 4.3 Certificate examination. Certificates of compliance, certified test reports, approval labels or listings for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Tests.

4.4.1 Sturdiness test. A completely assembled delivery and storage rack shall be loaded with an evenly spaced load of 600 pounds (272 kg) for the size 2 rack, 800 pounds (363 kg) for the size 4 rack, or 150 pounds (68 kg) on each shelf, (excluding the drop leaf) for the size 5 rack, as applicable, and shall be pushed ten times over a 3-inch (76 mm) wide floor obstacle, 1/4 inch (6.4 mm) high having 90-degree faces, with longitudinal axis of the obstacle set at an angle of 30 degrees to the direction of travel of the delivery and storage rack. The rack, including casters, shall be examined for permanent deformation to determine compliance with the sturdiness requirement of 3.4.

* 4.4.2 Load test. The racks as loaded in 4.4.1 shall be allowed to stand for 24 hours on a concrete floor or other type hard surface. The weights shall then be removed and shelves examined for any permanent deformation greater than 1/8 inch (3.2 mm).

* 4.4.3 Drop leaf test. A uniform 50 pound (23 kg) load shall be placed on the shelf and allowed to remain for 5 minutes. The load shall not extend beyond the borders of the shelf. The load shall be removed and shelf examined for requirements specified in 3.4.1.2.

5. PACKAGING

5.1 Levels A, B, and C. Preservation, packaging, packing, and marking shall be in accordance with the applicable requirements of MIL-B-2316 (see 6.2).

6. NOTES

* 6.1 Intended use. The delivery and storage racks covered by this specification are intended for use in moving, delivering, and handling meat and produce in commissary stores. The "V-top" configuration is intended to hold produce cases or crates for easy access and viewing.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size of delivery and storage racks required (see 1.2).
(c) Selection of the applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.4 Bracket. Suggested source is Stanley No. 794 B.G. 8 x 8 folding bracket or equal.

6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-R-43567A(GL) RACKS, DELIVERY AND STORAGE FOOD CONTAINER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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